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[14EE43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

POWER ELECTRONICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least **TWO** Questions from each Part.
2. Missing data may be suitably assumed.

PART - A

1. a. Mention and explain the different types of Power Electronic Converter Systems. Draw their input-output characteristics. **08**

b. What are the peripheral effects of Power Converter System? **06**

2. a. Give the structures, static characteristics and applications of IGBT **06**

b. The bi-polar transistor shown in figure 2(b) is specified to have β in the range of 8 to 40. The load resistance R_C=11Ω. The dc supply voltage V_{CC}=200 Volts and the input voltage to the base circuit V_B=10 Volts. If $V_{CE(Sat)}$=0.2Volt and $V_{BE(Sat)}$=0.7 Volts . Find ; **08**

i. Value of R_B that results in saturation with overdrive factor of 5.

ii. Forced β_f

iii. Power loss in the transistor.

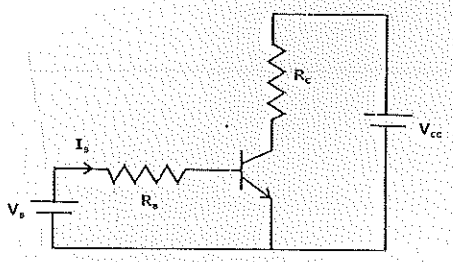

Figure 2(B)

3. a. Discuss the need of protection against di/dt and dv/dt. Explain how it is achieved with suitable circuit diagram. **08**

b. Ten thyristors are used in a string to withstand a dc voltage of V_S=15kV. The maximum leakage current and recovery charge differences of thyristors are 10mA and 150μc respectively. Each thyristor has a voltage sharing resistance of R=56kΩ and capacitance c_1=0.5μf. Determine ; **06**

i. Maximum steady state voltage sharing $V_{DS(max)}$.

ii. Steady state voltage derating factor.

iii. Maximum transient voltage sharing $V_{DT(max)}$.

4. a. What is meant by commutation? Distinguish clearly between natural and forced commutation. **06**

b. With necessary circuit diagram and waveforms, explain resonant pulse commutation. 08

PART – B

5. a. With the help of circuit diagram and waveforms, explain the principle of ON-OFF control in a single phase full-wave ac voltage controller with R load. Derive an expression for rms value of its output voltage. 08

b. For an ac voltage controller shown in figure 5(b), the delay angles of the thyristors T_1 and T_2 are equal, $\alpha_1=\alpha_2=2\pi/3$. Determine: 06

i. RMS output voltage

ii. Input power factor.

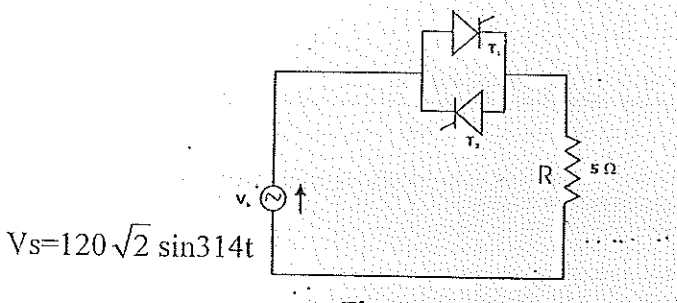

Figure 5(b)

6. a. With necessary circuit diagram and waveforms, explain the operation of Three-Phase half wave converter. 06

b. A single phase semi converter is operated from 120V, 50Hz ac supply. The load resistance is 10Ω. If the average output voltage is 25% of maximum possible average output voltage, determine; 08

i. Firing angle

ii. RMS and average output current

iii. RMS and average thyristor current

7. a. With the help of circuit diagram, explain the operation of Step-up chopper. Obtain an expression for average output voltage in terms of duty ratio. 06

b. A Step-down chopper has a resistive load of 10Ω and input voltage of V_S=220V. When the converter switch remains ON its voltage drop is 2 Volts and the chopping frequency f=1kHz. If the duty cycle is 50% determine; 08

i. Average output voltage

ii. RMS output voltage

iii. Chopper efficiency

iv. Effective input resistance of chopper.

8. a. With the help of a circuit diagram and waveforms, explain the operation of Single phase bridge inverter. 06

b. Explain with waveforms, how voltage is controlled in single phase inverter by sinusoidal and multiple pulse width modulation techniques. 08